

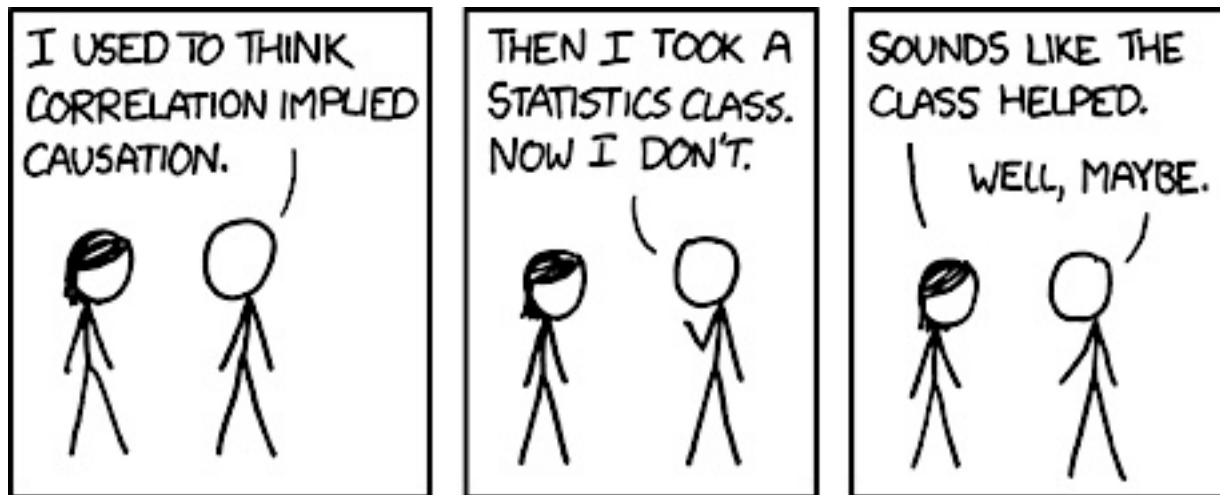
Sydney Informatics Hub

Core Research Facilities



THE UNIVERSITY OF
SYDNEY

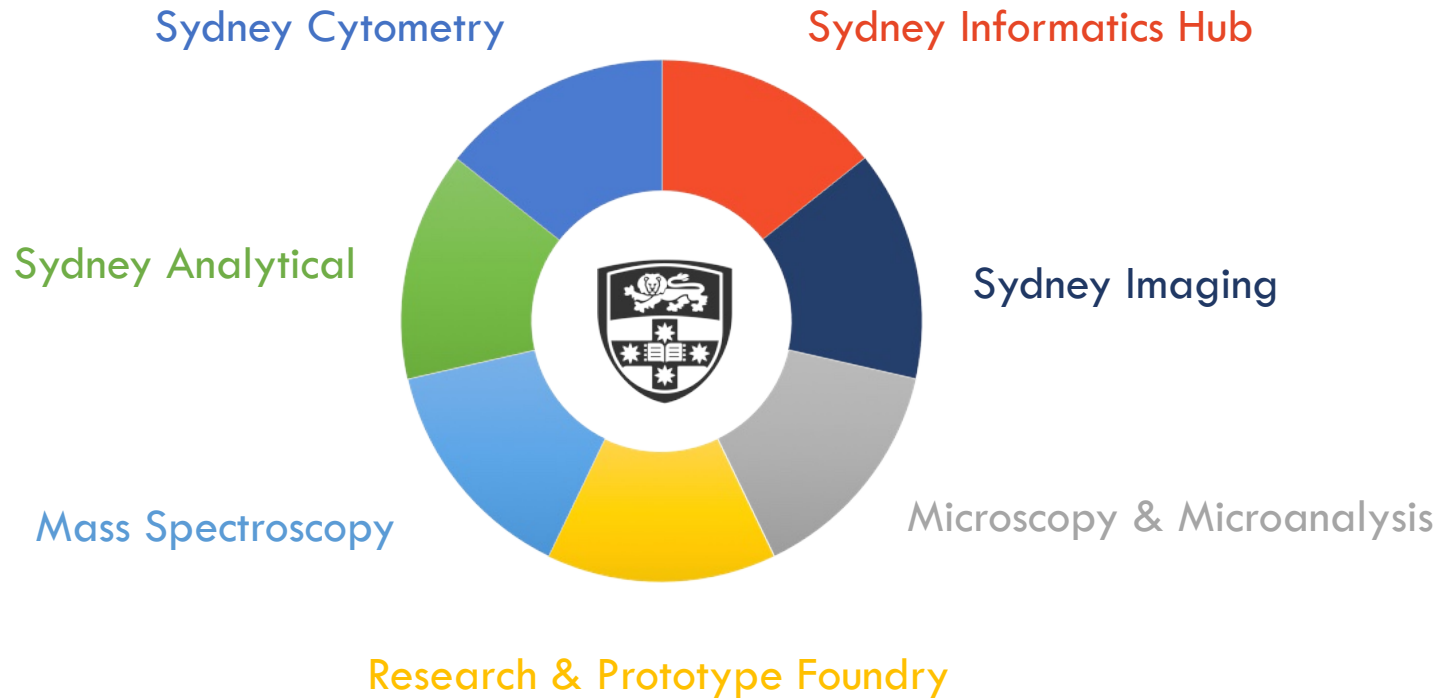
—
Sydney
Informatics Hub



Data science.



Core Research Facilities



Core Research Facilities

Sydney Cytometry

Sydney Informatics Hub



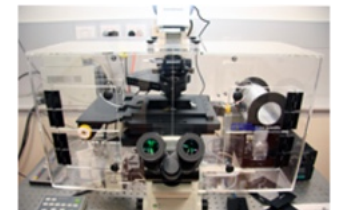
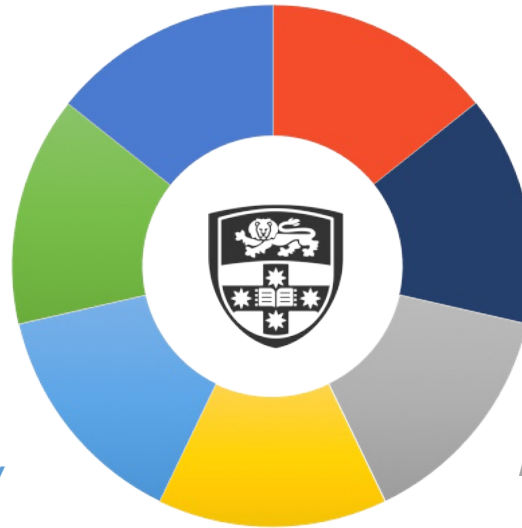
Sydney Analytical

Sydney Imaging

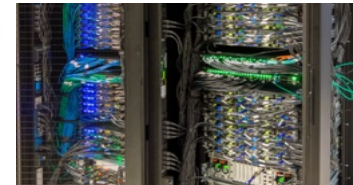
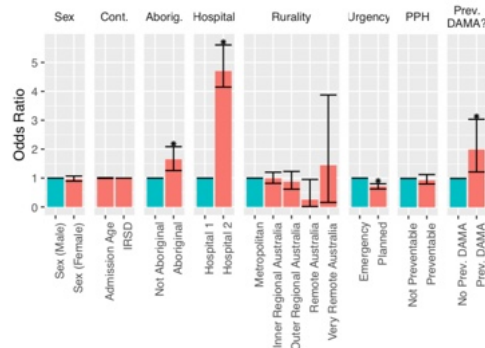
Mass Spectroscopy

Microscopy & Microanalysis

Research & Prototype Foundry



Core Research Facilities

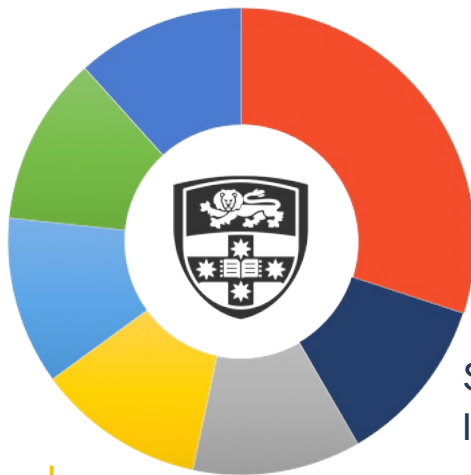


Sydney Informatics Hub

Cytometry

Analytical

Mass Spec



Sydney
Imaging

Prototype Foundry

Microscopy + Microanalysis

Training

Research Computing

Data management

Statistics

Data science & Informatics

Modeling, simulation and visualisation

Data community

informatics.sydney.edu.au

sih.info@sydney.edu.au

Sydney Informatics hub

Research computing,
bioinformatics, (big
data/big compute),
visualisation

Statistics consulting

Data science and
software engineering

Research data
management

Sydney Informatics hub

Research computing,
bioinformatics, (big
data/big compute),
visualisation

Statistics consulting

Training
Consultancy
Projects
Tools

Data science and
software engineering

Research data
management

Sydney Informatics hub

Services directly available to all HDR students!

Research computing,
bioinformatics, (big
data/big compute),
visualisation

Statistics consulting

Training
Consultancy
Projects
Tools

Data science and
software engineering

(Available through PIs)

Research data
management

Highlight: Training

2018 Upcoming Events and Training Sessions

A number of training sessions have been scheduled – click the links to register your attendance

Date	Training Course	Location	Signup
19 Sep	Regular expressions	Camperdown	Info Register
19 Sep	Version control with Git	Camperdown	Info Register
19 Sep	Hacky Hour	Camperdown	Info
20 Sep	Surveys in REDCap	Camperdown	Info Register
27 Sep	Advanced REDCap	Camperdown	Info Register
28 Sep	Accelerated Computing for Innovation	Camperdown	Info Register
3 Oct	Introduction to REDCap	Camperdown	Info Register
5 Oct	Introduction to Unix	Camperdown	Info Register
12 Oct	Intermediate-Advanced R	Camperdown	Info Register
15 Oct	Surveys in REDCap	Camperdown	Info Register
17 Oct	Introduction to Python	Camperdown	Info Register
23 Oct	Galaxy Australia – RNA-Seq	Camperdown	Info Register
23 Oct	EMBL-ABR: Using the UCSC Genome Browser	WEBINAR	Info Register
24 Oct	Introduction to MATLAB	Camperdown	Info Register
24 Oct	EMBL-ABR: A short tour of UCSC data	WEBINAR	Info Register
26 Oct	Intro to Artemis	Camperdown	Info Register
26 Oct	Intro to RDS and Data Transfer	Camperdown	Info Register
2 Nov	Introduction to R	Camperdown	Info Register
2 Nov	Next Generation Sequencing Analysis on Artemis	Camperdown	Info Register

- HPC
 - *Artemis, Pawsey*
 - *‘X on Artemis’*
- Programming
 - *Python, R, Unix*
- Data science & Statistics
 - *** Watch this space! ***
 - *Experimental/survey design*
- Bioinformatics
 - *SIH, EMBL partners*
- Research Data
 - *REDCap, RDS*

<https://informatics.sydney.edu.au/services/training/>

SIH: Research Computing

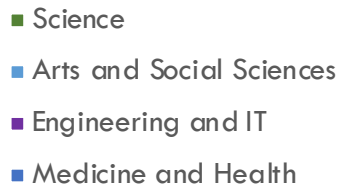
Research computing

HPC, Virtual Desktops,
Bioinformatics,
Modeling & Visualisation,
Data community

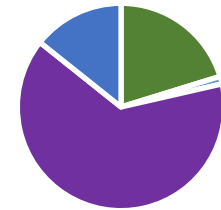
SIH: Research Computing

Research computing
HPC, Virtual Desktops,
Bioinformatics,
Modeling & Visualisation,
Data community

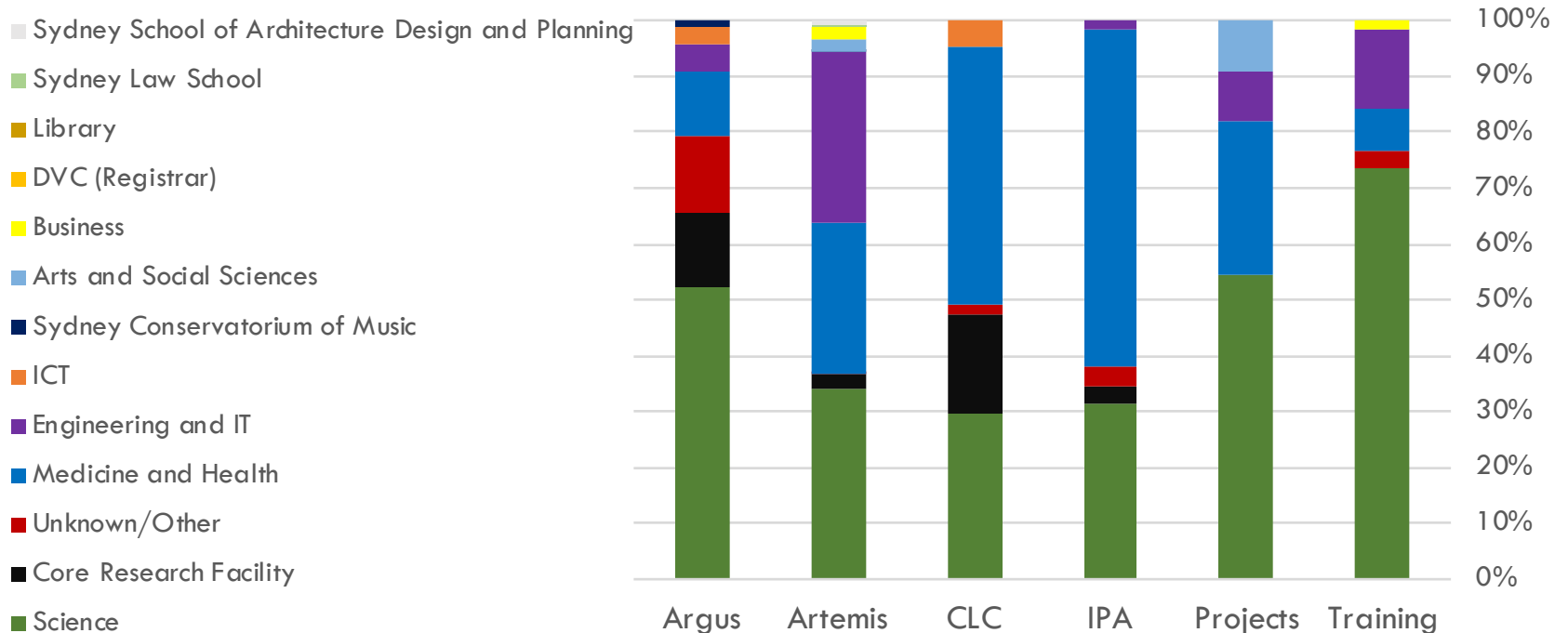
New Projects



New Artemis Users



Faculty affiliation of service users - August 2018



SIH: Research Computing

Research computing

HPC, Virtual Desktops,
Bioinformatics,
Modeling & Visualisation,
Data community

Cluster computing

Artemis HPC

Matlab DCS
MATLAB

CLC Genomics Server
QIAGEN



Argus VRD



Characterisation Virtual Laboratory
Genomics Virtual Laboratory



CLC Genomics Workbench
Ingenuity Pathways Analysis
QIAGEN

Genomics software

RDS



Virtual Desktops & Labs

Data storage

SIH: Research Computing

Research computing
HPC, Virtual Desktops,
Bioinformatics,
Modeling & Visualisation,
Data community



Data science community

The Sydney Informatics Hub presents

 THE UNIVERSITY OF SYDNEY
Core Research Facilities
Sydney Informatics Hub

**Accelerated Computing
for Innovation
Conference**

Friday 28 September 2018

Join us to celebrate innovation driven by cutting-edge tech, big data, machine learning and AI. Listen to fascinating talks from

Location
ABS Lecture Theatre 1130
Abercrombie Building (H70)
cnr Abercrombie St and Codrington St
University of Sydney



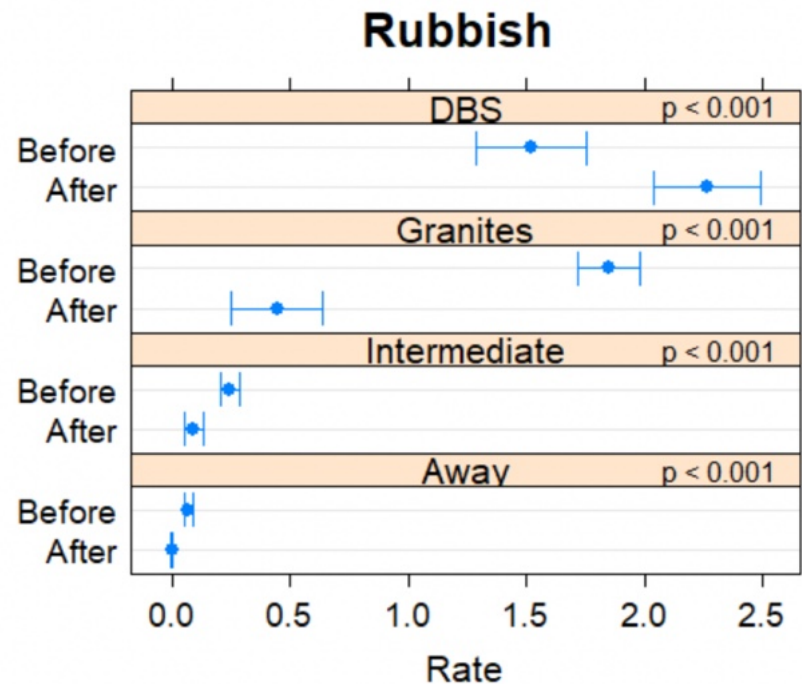
Who uses SIH services?

Who uses SIH services?

Statistical Consulting

Newsome Lab – SoLES – Species' responses to human-induced change

- Needed to assess changes in Dingo diet from analyses of their scat.
- SIH Stats Lead Chris Howden worked with client to **develop an experimental and model design.**
- Fit Poisson models to rates of incidence of various components in Dingo scat, before & after intervention



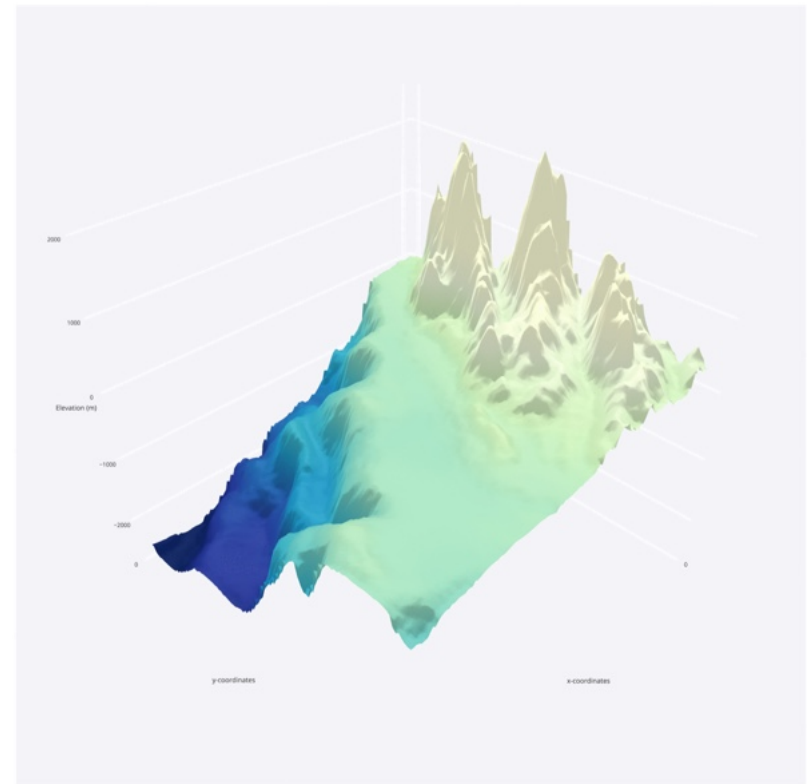
Source: Tom Newsome and SIH

Who uses SIH services?

Artemis High Performance Computing

EarthByte – Geoscience – Deep-time landscape evolution

- Researchers developed a Bayesian model for predicting how land eroded over time and climate
- Needed to **explore large parameter space** - unfeasible on single machines
- SIH Informatics Tech adapted the project's code to **run 100,000's of initial conditions** in parallel

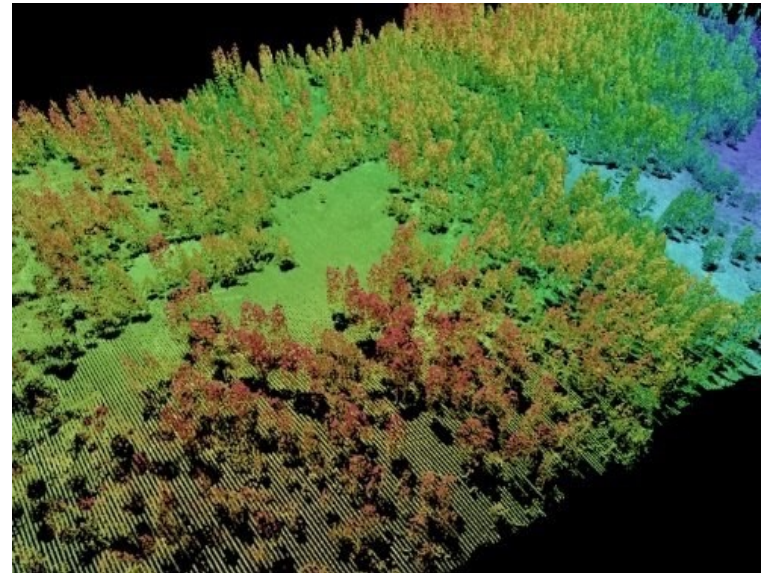


Who uses SIH services?

Argus Virtual Research Desktops

Sydney Institute for Agriculture – Vegetation change monitoring

- High resolution imaging of vegetation cover for conservation and agriculture.
- Needed a powerful **GPU**-enabled system to process and render ArcGIS and Pix4D **visualisations**
- Images captured by drones to be **uploaded directly to server**
- Geographically distant researchers all needed to update and work with data from **different locations**





NORMAL DISTRIBUTION



PARANORMAL DISTRIBUTION

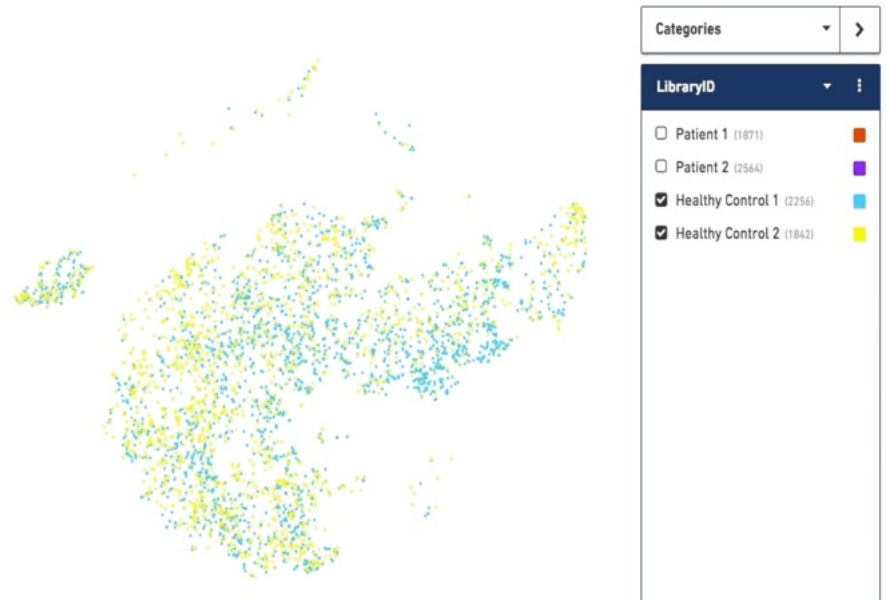
Franco.

Who uses SIH services?

Bioinformatics Consulting

Prof. Benjamin Tang – Medicine & Health – Influenza Research

- New equipment allowed the sequencing of individual immune cells
- Clinical researchers with **very little computing experience**, wanted to know if cells were different between patients with and without sepsis
- Bioinformatics Tech used **DNA sequencing on Artemis to classify immune cell populations**, identifying cell types unique to the sick patients

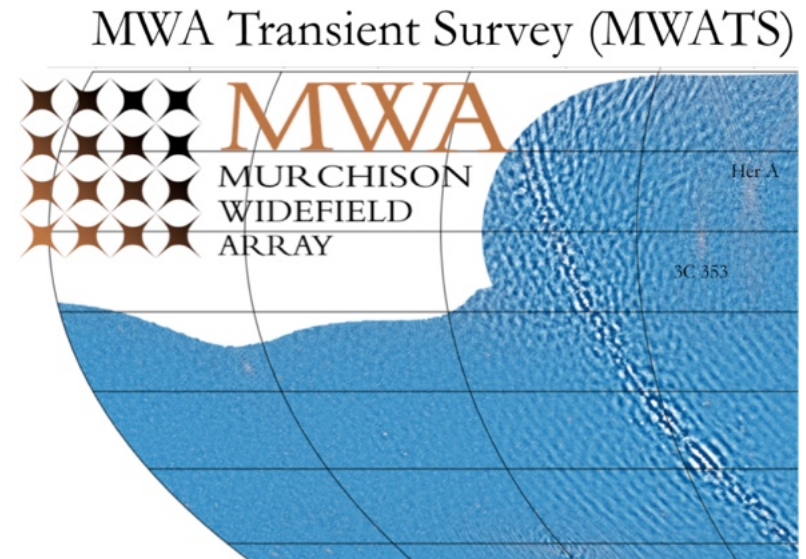


Who uses SIH services?

Data science and Research engineering

Prof Tara Murphy – Physics – “Scaling the Sky”

- Researchers wanted to look at the brightness of supermassive black holes over time
- This required tracking individual objects **across thousands of images**.
- Their software did not **scale up**, so an SIH engineer **optimised the database**, enabling analysis of 300,000 black holes.



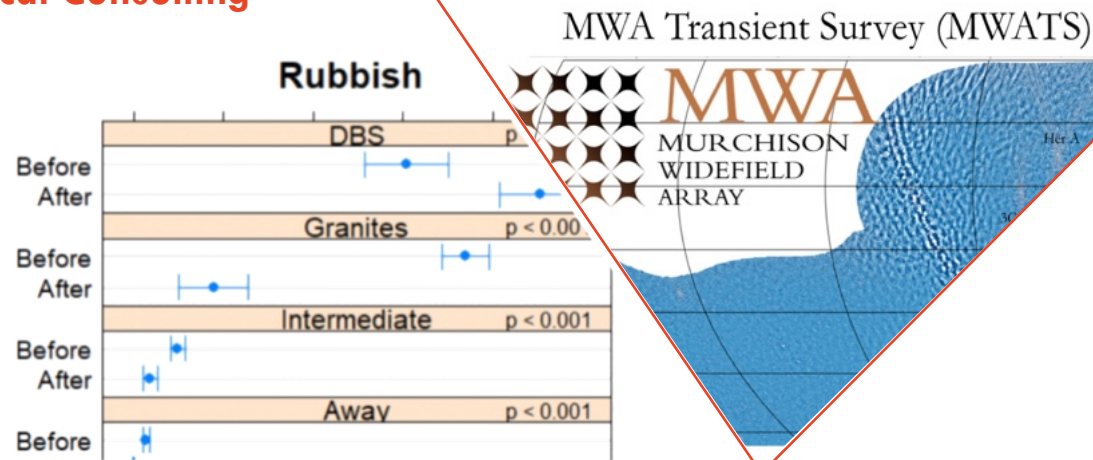
Source: M.Bell, T. Murphy, D. Kaplan, on behalf of the MWA

Who uses SIH services?

**Data Science &
Research Engineering**

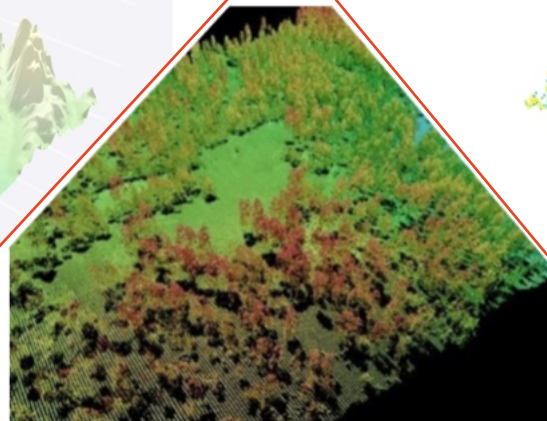
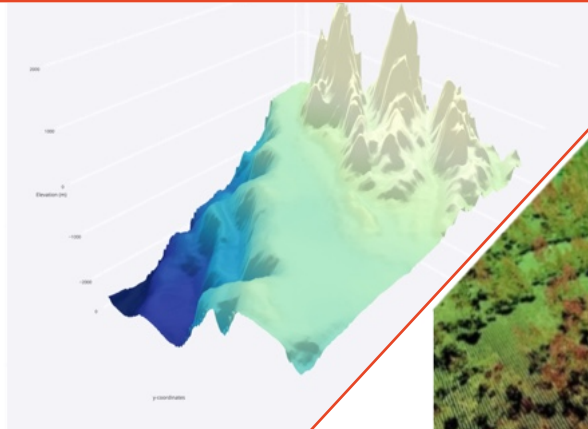
Statistical Consulting

**Research Data
Management**



(up next)

Artemis HPC



Argus VRD



Bioinformatics

Funding Opportunities

Core Research Facilities Access Scheme

- Funding for up to \$50,000 of SIH services (dedicated research computing, consultancy)

sydney.edu.au/research/facilities/user-access-scheme.html

SIH Publication Incentive

- Publications acknowledging SIH services can win up to \$1000
 - *The authors acknowledge the facilities, and the scientific and technical assistance of the Sydney Informatics Hub at the University of Sydney.*
 - *The authors acknowledge the Sydney Informatics Hub and use of the University of Sydney's high performance computing cluster, Artemis, for their contributions to the results reported herein.*

Contact Us!

Sydney Informatics Hub

W: informatics.sydney.edu.au

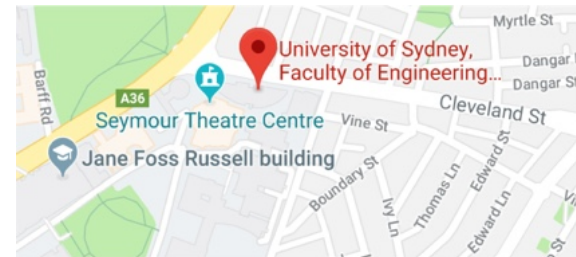
E: sih.info@sydney.edu.au

sih.training@sydney.edu.au

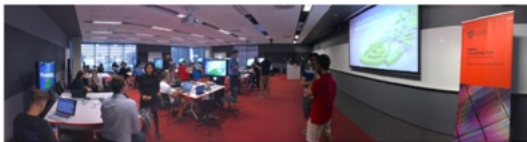
digital.research@sydney.edu.au

T: 8627 6286 (RC)

L: 5w, J12 – but about to move!



Training



- [SIH training schedule](#)
- [SIH course descriptions](#)
 - [Course slides and material](#)
- [Further training opportunities](#)

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informatics.sydney.edu.au/training

SIH Assistance Form

Please fill in details for the main contact for the issue below

*Required

Contact details

First Name *
Your answer

Last Name *
Your answer

informatics.sydney.edu.au/help

Thanks for listening!

NEXT:

**Research Data
Management!**

